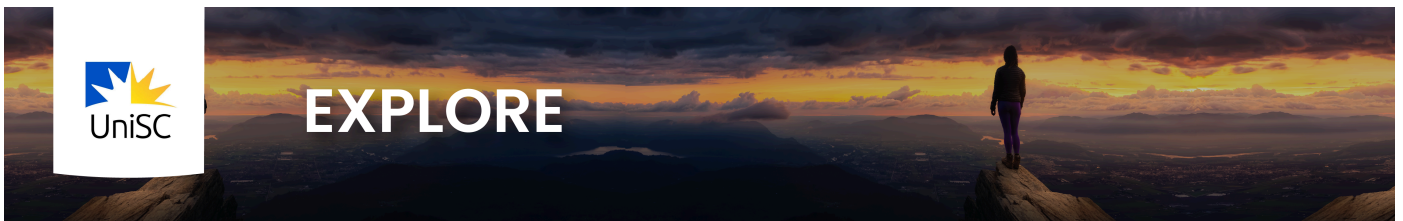


5.1 Introduction to Futures Contracts



Guiding Questions

1. Why do futures markets exist as a response to price risk in business and finance?
2. How does contract standardization improve liquidity and tradability?
3. How do initial margin, variation margin, and marking-to-market support market stability?
4. What trade-offs do firms face when hedging with futures contracts?



What will you learn?

- Explain the purpose of futures contracts in managing price uncertainty.
- Describe the main features of a futures contract, including standardisation, delivery terms, and exchange trading.
- Explain how a clearinghouse functions as a central counterparty through novation and daily settlement.
- Assess the benefits and limitations of hedging, including opportunity cost, liquidity pressure, and basis risk.

Why Futures Contracts Exist: The Problem of Price Risk

Every business that relies on a commodity, an interest rate, or a currency faces constant uncertainty about future prices. This unpredictability can make planning and budgeting

challenging.

For example, consider an airline that buys millions of litres of jet fuel every month. If global oil prices spike unexpectedly, the airline's fuel bill increases dramatically, which can erase profits overnight. Similarly, a wheat farmer planting crops in March has no idea what the selling price will be in October.

This exposure to price risk makes financial planning and investment difficult. Companies cannot sign long-term contracts or secure bank loans without some assurance of revenue stability.

To solve this, financial markets created futures contracts: formal agreements that allow a firm to lock in a price today for a transaction that will occur in the future.

The goal of a futures contract is not to generate profit, but to protect a known margin or cost level so the business can operate securely and focus on performance rather than speculation.

5.1.1 What is a Futures Contract?

A futures contract is a standardised, legally binding agreement to buy or sell an underlying asset at a specific price on a predefined future date.

The asset can be a commodity (like oil, wheat, or copper) or a financial instrument (like a stock index, exchange rate, or bond yield). These contracts are traded on organized futures exchanges, which operate under strict rules to ensure fairness and financial security.

Example

Suppose you run a bakery that uses 10 tonnes of flour each month. The price of wheat has been unstable, sometimes jumping 20% in a few weeks. To protect yourself, you buy a futures contract in March to purchase wheat at \$400 per tonne for delivery in June. Whatever happens to the market, your effective price per tonne will stay at \$400.

Thus, your cost becomes predictable, even if the global wheat market becomes volatile.

5.1.2 The Structure and Operation of Futures Contracts

Standardisation

Every futures contract traded on an exchange is standardized. This means all terms except for price are fixed in advance by the exchange:

Contract size: Specifies the quantity covered (e.g., one gold futures contract = 100 ounces).

Quality or grade: Defines product specifications (e.g., the grade of crude oil or type of wheat).

Delivery month: Specifies when the contract expires (March, June, September, December are common).

Delivery location: States where physical delivery could occur, even if most contracts are closed financially before expiry.

Because all these attributes are fixed, any trader can buy or sell the same contract with anyone else easily. This uniformity gives futures markets deep liquidity, enabling positions to be opened or closed instantly.

5.1.3 Clearinghouse – The Core of Market Integrity

Perhaps the most vital pillar of the futures market is the Clearinghouse. The clearinghouse acts as the central counterparty to all trades on the exchange.

When two traders agree to a futures contract—say, a seller (short) and a buyer (long)—they do not actually owe each other directly. Instead, the clearinghouse legally steps in between and becomes:

The buyer to every seller, and

The seller to every buyer.

This process, called novation, eliminates the fear that one party might default.

The clearinghouse also manages all deposits (called margins) and performs the daily marking-to-market process.

Real-World Clearinghouse Examples

CME Clearing at the Chicago Mercantile Exchange (USA).

ASX Clear (Futures) in Australia, part of the ASX 24 derivatives platform.

[ASX Clear \(Futures\) Information](https://www.asx.com.au/markets/clearing-and-settlement-services/asx-clear-futures) → [_\(<https://www.asx.com.au/markets/clearing-and-settlement-services/asx-clear-futures>\)](https://www.asx.com.au/markets/clearing-and-settlement-services/asx-clear-futures)

These institutions are globally recognized for maintaining financial stability and transparency in derivatives trading.

Analogy: Think of the clearinghouse as the trusted referee in a large, complex marketplace. It monitors every trade, holds deposits from both sides, and ensures that gains and losses are settled immediately and fairly every single day.

5.1.4 Margin Accounts

A margin is simply a safety deposit, not a payment for the commodity itself. It ensures both sides have enough funds to cover potential losses.

Initial Margin (IM): The amount that must be deposited to open a futures position—typically 5–10% of contract value.

Maintenance Margin: The minimum balance that must be maintained. If your account balance falls below this level, due to market losses, you will receive a margin call requesting more funds.

Variation Margin: The additional funds deposited to restore the account back to the initial margin level after a call.

Example: Imagine a gold futures contract worth \$200,000 requires an initial margin of 10% (\$20,000). If the market moves against you and your account balance drops to \$17,000 while the maintenance margin is \$18,000, you receive a margin call for \$3,000 to restore your balance.

Margins make the futures market credit-risk free, because all losses are settled daily—no build-up of unpaid debts occurs.

5.1.5 Marking-to-Market – The Daily Settlement Process

Marking-to-market is the daily process of adjusting all open futures positions to reflect the current market price.

If the price moves in your favour, your margin account is credited that day. If it moves against you, your account is debited instantly.

Example of Marking-to-Market: You sell one wheat futures contract for \$600 per tonne (contract size = 100 tonnes).

Day	Settlement Price	Daily Change	Profit/Loss	Margin Account Balance
0	600	–	–	\$5,000
1	590	-10	+\$1,000	\$6,000 (the idea is that you actually do not have the wheat but as market price has dropped, you can buy it at the lower market price from the market and sell it to the contract buyer (who agreed to buy from you at \$600/tonne) making a profit
2	610	+20	-\$2,000	\$4,000
3	605	-5	+\$500	\$4,500

Each day's difference is credited or debited in cash. If the balance falls below the maintenance level, the trader must deposit additional funds.

This mechanism ensures that losses are paid for immediately, protecting the clearinghouse and all other participants from default contagion.

5.1.6 Understanding the Risk Structure of Futures

Opportunity Cost: By locking in a future price, a trader gives up the opportunity to benefit if market prices later move favourably.

Example: A cotton producer sells futures at \$600, but the market price later rises to \$750. The producer must still deliver at \$600, missing the opportunity for the extra \$150 per bale.

Liquidity Risk (Margin Call Pressure): A hedge can succeed in theory but still create real-world cash flow stress. If market prices rise sharply against a short hedger, they must post large daily margin payments, even if their physical sale months later offsets those losses. This can strain liquidity.

Basis Risk: The basis is the difference between spot () and futures prices ():

If the spot and futures prices do not move perfectly together, the hedge will not completely offset the exposure. This imperfect correlation is called basis risk.



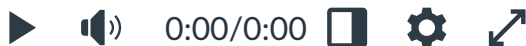
Watch

Futures contracts: The main idea



Details

Transcript



You now understand what functions are and why analysts rely on them.

Next, you'll **practice using and modifying functions** and take your first step towards writing your own (a key milestone in developing analytical confidence).

5.1.7 Hedging Equities with Futures (Locking in a Price)

The same risk-management principles that apply to physical commodities like cotton apply to financial equities. Whether an investor wants to protect an individual company stock or a broad market basket like the S&P 500, they can use derivatives to manage price risk.

An investor who currently owns an equity position and plans to liquidate it in the future can sell (short) a futures contract today to eliminate price volatility.

- **Single Stock:** To hedge an individual stock, the investor sells a **Single Stock Future (SSF)** contract on that specific ticker.
- **S&P 500:** To hedge a diversified portfolio that mirrors the broader market, the investor sells **S&P 500 index futures**.
-

The Mechanics & Trade-off:

Just like the cotton producer who locks in a selling price, the equity investor creates a symmetric hedge. No upfront cash premium is paid to establish the position. If the stock or the S&P 500 drops in value, the gains from the short futures contract perfectly offset the losses on the shares. However, if the market rallies, the losses on the futures contract completely wipe out the equity gains. The investor achieves absolute certainty over their final exit price but completely surrenders any potential upside.

Worked Example – The Cotton Producer:

Setup

- Producer: A local cotton farmer
- Quantity: 2,000 bales
- Initial futures price: \$600/bale
- Target net revenue: $2,000 \times \$600 = \$1,200,000$

In March, the farmer sells futures contracts locking in \$600 per bale for October delivery. Whatever happens to the cotton market between March and October, the farmer's net revenue will remain at \$1,200,000.

Case A – Market Price Falls

At delivery: spot price = \$500/bale

Physical sale = $2,000 \times \$500 = \$1,000,000$

Futures profit = $2,000 \times (\$600 - \$500) = \$200,000$

Net revenue = $\$1,000,000 + \$200,000 = \mathbf{\$1,200,000}$

The farmer sold cotton at a disappointing market price of \$500. However, the futures position generated a \$200,000 profit, because the farmer had locked in the right to sell at \$600 when the market is only \$500. The two figures combine to deliver exactly the target revenue.

Case B – Market Price Rises

At delivery: spot price = \$700/bale

Physical sale = $2,000 \times \$700 = \$1,400,000$

Futures loss = $2,000 \times (\$700 - \$600) = \$200,000$

Net revenue = $\$1,400,000 - \$200,000 = \mathbf{\$1,200,000}$

The farmer sold physical cotton at a favourable price of \$700. However, the futures position generated a \$200,000 loss, because the farmer had agreed to sell at \$600 when the market is now \$700; the buyer of the futures contract captures that \$100 difference per bale. The net result is again \$1,200,000.

The hedge worked as intended, but it came at a cost: the farmer gave up \$200,000 of upside that would have been captured without the hedge.

Key Insight

In both cases, net revenue equals \$1,200,000. The hedge achieved price certainty. However, certainty has a cost: the farmer neither benefits from a price rise nor suffers from a price fall. The hedge converts a variable outcome into a fixed one, which is precisely its purpose.

Example of the Clearinghouse System in Practice:

Suppose the same cotton producer sells one futures contract on Australia's ASX 24 Exchange, cleared by ASX Clear (Futures).

The cotton producer sells a contract through a broker at \$600 and posts the required initial margin.

A textile mill buys that contract at the same price and posts its own margin.

ASX Clear becomes a counterparty to both traders:

The buyer owes ASX Clear, not the seller.

The seller owes ASX Clear, not the buyer.

Each day, ASX Clear marks all contracts to market. Profits flow from one side's margin account to the other automatically.

Even if one party defaults or disappears, ASX Clear's margin reserves and financial safeguards guarantee that no loss will spread through the system.

This system has been operating reliably for decades and is the cornerstone of why global futures markets are considered among the safest and most liquid financial platforms.

More information: [ASX Clear \(Futures\) – Clearing and Settlement Mechanism](https://www.asx.com.au/markets/clearing-and-settlement-services/asx-clear-futures) 
(<https://www.asx.com.au/markets/clearing-and-settlement-services/asx-clear-futures>)

Summary – The Economics of Price Certainty

Futures Contracts are formal agreements that guarantee a future buying or selling price, transforming uncertainty into certainty.

Standardization ensures contracts are uniform and tradable by anyone.

The Clearinghouse (e.g., CME Clearing, ASX Clear) guarantees performance, standing between buyers and sellers.

Margins and Daily Marking-to-Market make futures markets almost free from default risk, but they introduce the need for continuous liquidity.

Hedging provides revenue or cost stability, not speculative profit.

The key trade-off: Price protection versus opportunity cost. Hedgers limit both gains and losses but enable financial predictability.

Useful Reading Links:

[CME Group Education Portal – Futures Basics](https://www.cmegroup.com/education.html)  (<https://www.cmegroup.com/education.html>)

[ASX Education Centre – Derivatives Learning Materials](https://www2.asx.com.au/markets/trade-our-derivatives-market/education) 
(<https://www2.asx.com.au/markets/trade-our-derivatives-market/education>)

[U.S. Commodity Futures Trading Commission \(CFTC\) – Futures Guide](https://www.cftc.gov/LearnAndProtect/EducationCenter/index.htm) 
(<https://www.cftc.gov/LearnAndProtect/EducationCenter/index.htm>)

Reflect

- Why does the clearinghouse eliminate counterparty risk?
 - Explain why futures contracts are marked-to-market daily
 - Describe how a margin call protects the system
 - What is basis risk, and why does it matter?
-

Solve

Question 1

A gold miner sells gold futures at \$4,800/oz and the price rises to \$5,100/oz. What is the loss per ounce?

Question 2

You short 5 oil futures contracts (1,000 barrels each) at \$90. The price then rises to \$94. Calculate the total dollar gain or loss on this position, and state whether it is a gain or a loss for the trader.

Question 3

Initial margin = \$12,000 per contract; maintenance = \$9,500. After a daily loss, the account balance falls to \$9,000. Calculate the margin call.

Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3293288?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3293288?wrap=1).

Critically evaluate

- Should a firm that wants to maximize profits use futures hedging?
- Name industries where futures hedging is essential.